GRAPHIC ERA HILL UNIVERSITY
DEHRADUN CAMPUS



PROJECT REPORT
"Smart Greenhouse IoT System for Real-Time Monitoring and Alert"
MINI PROJECT
B.Tech (CSE)

SEMESTER-III

SUMBITTED TO:
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GRAPHIC ERA HILL UNIVERSITY DEHRADUN

CERTIFICATE

Certified that AKHIL RAWAT (Roll no 2318259) has developed a mini project on "**Smart Greenhouse IoT System for Real-Time Monitoring and Alert**" in Graphic Era Hill University, Dehradun.

Class Co-ordinator

(Mr Shushant Chamoli)

Acknowledgement

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SERIAL NUMBER	TOPIC
1.	ABSTRACT
2.	PROBLEM STATEMENT
3.	AIM
4.	TOOLS AND TECNOLOGY USED
5.	METHODOLOGY
6.	IMPLEMENTATION
7.	SCOPE OF WORK
8.	RESULTS
9.	CONCLUSION

1. ABSTRACT

This project aims to create a smart plant monitoring system that simplifies plant care by providing real-time environmental and soil data. The system uses an ESP8266 microcontroller to collect data from sensors, including a DHT11 for temperature and humidity, a soil moisture sensor, an LDR for light intensity, and an ultrasonic sensor for proximity detection. The data is sent to the Blynk cloud via Wi-Fi for remote monitoring and control using mobile app. Visual and auditory alerts are triggered when soil moisture is low or when an object is detected near the plant. This system helps users monitor plant health efficiently and ensures timely care, making it an effective solution for smart gardening.

2. Problem Statement

Greenhouses require constant monitoring of environmental parameters such as temperature, humidity, soil moisture, light intensity, and proximity of objects or animals to ensure optimal plant growth. Manual monitoring is time-consuming and prone to errors. This project aims to develop an IoT-based system to automate the monitoring and control of these parameters in a greenhouse.

Motivation

The motivation behind this project is to leverage IoT technology to create a smart greenhouse system that can:

i. Automatically monitor critical environmental parameters.
ii. Provide real-time alerts and notifications to the user.
iii. Ensure optimal growing conditions for plants with minimal human intervention.
iv. Reduce resource wastage and improve crop yield.

3. AIM

To design and implement an IoT-enabled smart plant monitoring system that utilizes the ESP8266 microcontroller to collect, process, and transmit data from various sensors, including temperature, humidity, soil moisture, light intensity, and proximity detection sensors. The system aims to provide real-time monitoring and control of plant health through a mobile or desktop application using the Blynk cloud. It also aims to enhance plant care efficiency by generating automated alerts for low soil moisture and nearby object detection using an LED indicator and buzzer, ensuring optimal environmental conditions for plant growth.

4. Tools Used

Hardware:

i. NodeMCU (ESP8266): Microcontroller for IoT connectivity.
ii. DHT11 Sensor: Measures temperature and humidity.
iii. Soil Moisture Sensor: Detects the moisture level in the soil.
iv. LDR (Light Dependent Resistor): Measures light intensity.
v. Ultrasonic Sensor (HC-SR04): Detects the proximity of objects or animals near the plants.
vi. Buzzer: Provides audible alerts.
vii. LED: Indicates soil moisture alerts.

Software:

i. Arduino IDE: For programming the NodeMCU.
ii. Blynk IoT Platform: For real-time monitoring and notifications.
iii. Programming Language: C++ (Arduino)

5. Methodology

The system is designed to monitor and control the greenhouse environment using IoT. The methodology involves the following steps:

a. Sensor Integration

i. DHT11 Sensor: Measures temperature and humidity.
ii. Soil Moisture Sensor: Detects the moisture level in the soil.
iii. LDR: Measures light intensity.
iv. Ultrasonic Sensor: Detects the proximity of objects or animals near the plants.

b. Data Collection and Processing

i. Sensor data is collected and processed by the NodeMCU microcontroller.
ii. The data is sent to the Blynk IoT platform for real-time monitoring.

c. Alerts and Notifications

i. If soil moisture is below 60%, an LED is turned on, and a notification is sent.
ii. If an object or animal is detected within 30 cm of the plants, a buzzer is activated, and a notification is sent.
iii. If low light intensity is detected, a notification is sent.

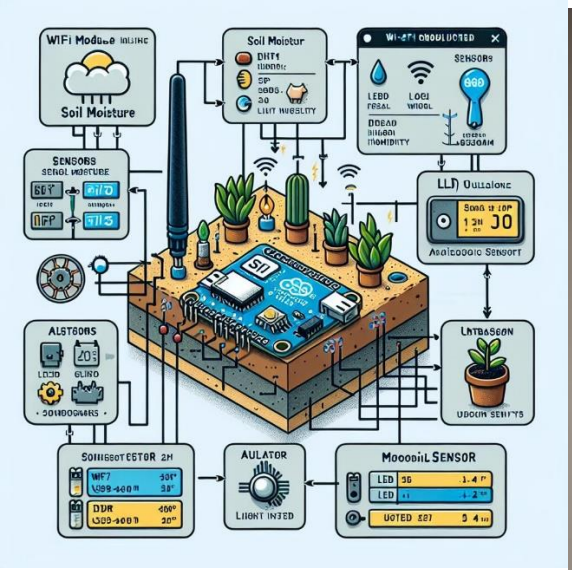
d. User Interface

The Blynk app provides a user-friendly interface to monitor sensor data and receive alerts.

6. Implementation

a. Circuit Diagram

The NodeMCU is connected to the DHT11, soil moisture sensor, LDR, ultrasonic sensor, buzzer, and LED as per the pin configuration in the code.





b. Code Overview

The code is written in the Arduino IDE and uses the Blynk library for IoT connectivity. Key functions include:

i. sendSensorData(): Collects and sends sensor data to Blynk.
ii. getDistance(): Calculates the distance using the ultrasonic sensor.
iii. sendNotification(): Sends alerts to the Blynk app.

c. Blynk App Setup

Virtual pins are configured in the Blynk app to display temperature, humidity, soil moisture, light intensity, and distance.

Notifications are set up to alert the user about critical conditions

Scope of Work

1. **System Design and Development**
    - Develop a hardware system integrating the ESP8266 microcontroller with sensors such as DHT11 (temperature and humidity), soil moisture sensor, LDR (light intensity), and ultrasonic sensor.
    - Interface the microcontroller with actuators like an LED and buzzer for visual and auditory alerts.
2. **IoT Integration**
    - Connect the system to the Blynk cloud platform using Wi-Fi for real-time data transmission.
    - Enable users to monitor plant health remotely through a mobile or desktop application.
3. **Data Processing and Alerts**
    - Collect and process sensor data to measure environmental and soil conditions.
    - Generate alerts for low soil moisture, low light intensity, or object proximity using the Blynk app, LED, and buzzer.
4. **Real-Time Monitoring and Notifications**
    - Display real-time data such as temperature, humidity, soil moisture percentage, light intensity, and distance on the Blynk app.
    - Notify users through push notifications to ensure timely plant care.
5. **Application and Usability**
    - Provide a user-friendly solution for plant monitoring and care in homes, offices, and gardens.
    - Enhance plant health management by ensuring optimal environmental conditions.

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7. Results

The system successfully monitors and controls the greenhouse environment. Key results include:

i. Real-time monitoring of temperature, humidity, soil moisture, light intensity, and proximity.
ii. Automatic alerts and notifications for critical conditions.
iii. Efficient control of environmental parameters to ensure optimal plant growth.

Precautions

i. Sensor Calibration: Ensuring accurate readings from the soil moisture sensor and LDR required calibration.
ii. Wi-Fi Connectivity: Initial challenges in connecting the NodeMCU to the Wi-Fi network were resolved by ensuring proper credentials and signal strength.
iii. Blynk Integration: Configuring the Blynk app and virtual pins required careful setup.

8. Conclusion

The Smart Greenhouse Monitoring and Control System Using IoT is an effective solution for automating greenhouse management. It provides real-time monitoring, alerts, and control of environmental parameters, ensuring optimal conditions for plant growth. Future improvements could include integrating additional sensors, implementing machine learning for predictive analysis, and expanding the system for larger-scale greenhouses.

References

i. Blynk IoT Platform Documentation: https://blynk.io/
ii. Arduino IDE Documentation: https://www.arduino.cc/
iii. DHT11 Sensor Datasheet
iv. NodeMCU ESP8266 Documentation